Decoding Parameter and Sampling Analysis Report

A Comparative Study of Temperature, Top-K, and Top-P Sampling

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Repository: github.com/Huzaifa150921/llm-Sampling-Analysis **Models:** Llama 3.1 (8B) & Qwen 2.5 (7B) **Framework:** Ollama

ABSTRACT

Large Language Models (LLMs) generate text by selecting tokens according to probability distributions governed by sampling parameters like temperature, Top-K, and Top-P. This project evaluates these parameters using two locally executed models: Llama 3.1 (8B) and Qwen 2.5 (7B). Evaluated across six sampling configurations covering JSON generation, arithmetic reasoning, factual QA, and creative writing, Qwen 2.5 achieved 100% JSON structure validity across all configurations, while Llama 3.1 scored 0% under the strict parsing pipeline. Low-temperature determinism reached 91.67% for Qwen 2.5 vs. 83.33% for Llama 3.1, while both models demonstrated 100% accuracy on basic arithmetic and factual keyword extraction tasks.

144	100%	0%	91.67%
Total Prompt Runs	Qwen JSON Validity	Llama JSON Validity	Peak Determinism

1. INTRODUCTION

Generative Large Language Models are widely applied across automated question answering, content synthesis, code generation, and structured data extraction. However, model outputs depend heavily on the specific decoding strategy applied during inference. Sampling parameters dynamically manipulate the token selection space, balancing output predictability against linguistic variety. Understanding these mechanics is essential for safety-critical and structured tasks that require valid schema output, deterministic reproducible results, or high factual fidelity. This report evaluates Llama 3.1 (8B) and Qwen 2.5 (7B) under controlled decoding environments.

2. PROJECT OBJECTIVES

- **Determinism Analysis:** Quantify output consistency across varying temperature, Top-K, and Top-P configurations.
- **Structural Validation:** Measure output adherence to pre-defined strict JSON structures.
- **Task Accuracy:** Benchmark simple arithmetic reasoning and factual recall using targeted evaluation proxies.
- **Latency Benchmarking:** Record and compare end-to-end response generation times across models and hardware settings.

3. BACKGROUND & DECODING MECHANICS

3.1 Temperature

Temperature rescales the logit distribution before softmax computation. Lower temperatures (e.g., 0.0) sharpen the distribution, favoring high-probability tokens and leading to greedy or near-deterministic decoding. Higher temperatures flatten the distribution, permitting lower-probability tokens and boosting output diversity.

3.2 Top-K Sampling

Top-K truncates the candidate token pool to the *K* most probable tokens. Setting K=10 strongly restricts candidate options, whereas K=80 provides a broader candidate space for higher linguistic variance.

3.3 Top-P (Nucleus) Sampling

Top-P selects dynamically from the smallest cumulative subset of tokens whose aggregated probability mass exceeds threshold *P*. Tested thresholds of 0.50 and 0.95 balance tight distribution clipping against broad vocabulary access.

3.4 Evaluation Metrics

Key metrics include *Determinism* (% exact normalized matches across repeated executions), *JSON Structure Validity* (strict schema parsing), *Arithmetic Accuracy*, and *Factual Keyword Accuracy*.

4. EXPERIMENTAL METHODOLOGY

The evaluation framework was developed using Python, leveraging Ollama for local model execution. Prompts were categorized across four distinct domains and executed three times under six discrete sampling configurations (144 total prompt invocations across both models).

5.1 Prompt Categories

CATEGORY	PROMPT OBJECTIVE	TARGET OUTPUT / VALIDATION RULE
JSON	Generate object with name, age, city	Valid JSON containing specified schema keys
Mathematics	Calculate arithmetic expression (15 + 8)	Numerical extraction matching exact target (23)
Factual	Identify Pakistan's capital & national language	Keyword detection ("Islamabad" and "Urdu")
Creative	Write a short story about a robot on Mars	Qualitative variation & determinism measure

5. LOW-TEMPERATURE BASELINE RESULTS (TEMP = 0.0)

Baseline evaluation conducted at Temperature = 0.0, Top-K = 40, Top-P = 0.9.

EVALUATION METRIC	LLAMA 3.1 (8B)	QWEN 2.5 (7B)	OBSERVED DELTA / NOTE
Determinism	83.33%	**91.67%**	Qwen showed superior repeatability at T=0.0
JSON Validity	0.00%	100.00%	Llama added conversational wrapping text
Arithmetic Accuracy	100.00%	100.00%	Both correctly evaluated 15 + 8 = 23
Factual Keyword Accuracy	100.00%	100.00%	Identified "Islamabad" and "Urdu" perfectly
Average Latency	10.96 s	**7.87 s**	Qwen was 28.2% faster on initial run

6. COMPREHENSIVE RESULTS ACROSS ALL SAMPLING CONFIGURATIONS

MODEL	PARAMETER TESTED	SETTING	DETERMINISM	JSON VALIDITY	ARITH. ACC.	FACT. ACC.	AVG. TIME
Llama 3.1 (8B)	Temperature	0.0	83.33%	0%	100%	100%	10.96 s
Llama 3.1 (8B)	Temperature	0.8	50.00%	0%	100%	100%	3.46 s
Llama 3.1 (8B)	Top-K	10	66.67%	0%	100%	100%	3.14 s
Llama 3.1 (8B)	Top-K	80	66.67%	0%	100%	100%	3.16 s
Llama 3.1 (8B)	Top-P	0.50	75.00%	0%	100%	100%	3.04 s
Llama 3.1 (8B)	Top-P	0.95	66.67%	0%	100%	100%	3.05 s
Qwen 2.5 (7B)	Temperature	0.0	91.67%	100%	100%	100%	7.87 s
Qwen 2.5 (7B)	Temperature	0.8	83.33%	100%	100%	100%	2.58 s
Qwen 2.5 (7B)	Top-K	10	83.33%	100%	100%	100%	2.49 s
Qwen 2.5 (7B)	Top-K	80	83.33%	100%	100%	100%	2.57 s
Qwen 2.5 (7B)	Top-P	0.50	83.33%	100%	100%	100%	2.52 s
Qwen 2.5 (7B)	Top-P	0.95	83.33%	100%	100%	100%	2.52 s

7. KEY FINDINGS & DISCUSSION

JSON Adherence & Formatting: Qwen 2.5 produced clean JSON wrapped in Markdown fences that the parser easily handled. Llama 3.1 routinely attached introductory prose (e.g., "Here is your requested JSON object:"), causing complete validation failure under strict full-string parsing. This emphasizes the impact of output parsing logic on raw model evaluations.

Determinism Sensitivity: Increasing temperature from 0.0 to 0.8 decreased determinism significantly for Llama 3.1 (83.33% → 50.00%) and moderately for Qwen 2.5 (91.67% → 83.33%), confirming theoretical expectations regarding logit flattening.

8. Limitations

- Small test suite (4 prompts, 3 repeats per setting).
- Basic keyword matching cannot detect nuanced hallucinations.
- Model warm-up affected T=0.0 latency benchmarks.

9. Future Work

- Implement schema-based JSON regex extraction.
- Expand parameters across fine-grained sweeps.
- Export automated benchmark logs to CSV format.

10. CONCLUSION

Decoding parameters substantially affect output predictability and structural compliance. Qwen 2.5 exhibited strong zero-shot JSON compliance and higher overall determinism. Careful selection of sampling settings—paired with robust post-processing—is vital for deploying LLMs in structured software pipelines.

References: [1] Repository: github.com/Huzaifa150921/llm-Sampling-Analysis | [2] Ollama Framework: docs.ollama.com | [3] Python & NumPy Evaluation Stack